

# Curriculum vitae

## PERSONAL DETAILS

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**Full Name:** Md Abdullah-Al Mamun

**Researcher ID:**  [ORCID](#)  [Google Scholar](#)  [Sciprofiles](#)  [ResearchGate](#)

**Online Profiles:**  [LinkedIn](#) |  [GitHub](#) | [Portfolio](#)

### Contact Details:

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**Phone number:** +358407384474 

**Date of CV:** April 2026

## DEGREES

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### Master of Science (M.Sc.) in Computing Sciences (Specialisation in Data Science)

Tampere University, Finland | **Completion Date:** 4 May 2024

#### Master's Thesis:

*Map out the best BI tool and SPC software for a case company: selection and rationalization process.*

**Supervisors:** Professor Martti Juhola (Tampere University), Timo Poranen (Lecturer, Tampere University), Henna Strömberg (Valmet Automotive EV Power Oy, Salo, Finland) | Published in [Trepo](#).

#### Relevant Coursework:

*Pattern Recognition and Machine Learning (5/5), Data Mining (5/5), Machine Learning Algorithms (5/5), Dimensionality Reduction and Visualization (4/5), Statistical Inference (4/5).*

### Bachelor of Science (B.Sc.) in Biotechnology and Genetic Engineering

Mawlana Bhashani University of Science and Technology, Bangladesh | **Completion Date:** 30 January 2010

#### Core Coursework:

*Molecular Biology, **Bioinformatics**, Biostatistics, Biochemistry, Genetics.*

### Bachelor of Science (B.Sc.) in Information Technology (Software Engineering Major)

Metropolia University of Applied Sciences, Finland | **Completion Date:** 18 December 2017

#### Relevant Coursework:

*Advanced Programming (Python, Java), Software Engineering, Web Development, Android Application Development.*

## TECHNICAL SKILLS

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Python, R, MATLAB, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, SQL, Git, Power BI, Excel.

## LANGUAGE SKILLS

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● **English:** Professional and Fluent ● **Finnish:** A2.2 (CEFR self-assessment) ● **Bengali:** Native

## PREVIOUS WORK EXPERIENCE

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### Master's Thesis worker

[Valmet Automotive EV power Oy](#)

**Salo, Finland**

**March 2022 – June 2023**

- Thesis Topic: “Map out the best BI tool and SPC software: selection and rationalization process.”
- Utilized Python, Power BI, and Excel to analyze and visualize data for evaluating and selecting the optimal Business Intelligence (BI) tool and Statistical Process Control (SPC) software.
- Recommended Microsoft Power BI based on comprehensive analysis, leading to its successful adoption by Valmet as the preferred BI tool.

### Laboratory and quality Operator

[Valmet Automotive EV power Oy](#)

**Salo, Finland**

**August 2021 – November 2023**

- Conducted quality checks for 48-volt battery packs, achieving a 98% quality rate.
- Analyzed production line data using Excel and visualized findings in Power BI, leading to improved decision-making.

- Focused on enhancing skills in data science and AI through independent research and ongoing collaboration with academic peers.
- Pursuing independent research collaborations and preparing manuscripts for peer-reviewed journals.

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## RESEARCH OUTPUT

### ● Published / Accepted (peer-reviewed): 8

1. **Mamun, M. A.**, Aktar, M. N., Mallick, J., Alqadhi, S., & Islam, A. R. M. T. (2026). Hybrid ensemble machine learning for groundwater level prediction in data-scarce northern Bangladesh. *Urban Water Journal*. <https://doi.org/10.1080/1573062X.2026.2649885>

**Contribution:** *Conceptualization, Writing – review & editing, Visualization, Software, Investigation, Data curation.*

2. Islam, A. R. M. T., **Mamun, M. A.**, Tasnuva, A., Aktar, M. N., Mishra, M., Al Mamun, A., & Moin, M. J. (2026). Living on the flood line: Constructing and validating a combined multidimensional resilience index for rural riverine floodplain communities. *Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2026.181581>

**Contribution:** *Writing – review & editing, Visualization, Software, Investigation, Data curation.*

3. Islam, A. R. M. T., **Mamun, M. A.**, Uddin, M. N., Sowrav, S. F. F., Alam, M. N. E., Khan, S. R., Chowdhury, M. H., & Choudhury, T. R. (2026). Explainable and physics-informed machine learning for seasonal water quality prediction in the monsoon-driven Padma River Basin, Bangladesh. *Environmental Geochemistry and Health*. <https://doi.org/10.1007/s10653-026-03031-z>

**Contribution:** *Writing – review & editing, Visualization, Software, Investigation, Data curation.*

4. **Mamun, M. A.**, Aktar, M. N., Uddin, M. N., Chowdhury, M. H., Islam, M. S., Rahman, M. S., Uddin, M. R., Hossain, M. J., Zahid, A., Senapathi, V., & Islam, A. R. M. T. (2026). Hybrid data-driven framework for interpretable prediction of nitrate and sulfate risks in coastal aquifers. *Science of the Total Environment*, 1011, 181190. <https://doi.org/10.1016/j.scitotenv.2025.181190>

**Contribution:** *Writing – original draft, Visualization, Software, Formal analysis, Conceptualization.*

5. Islam, A. R. M. T., Varol, M., Mallick, J., **Mamun, M. A.**, Mia, M. Y., Siddique, M. A. B., Islam, M. S., & Aktar, M. N. (2026). Tracing source footprints of heavy metal(oid)s in coastal soils using traditional statistical techniques and machine learning data-driven models. *Marine Pollution Bulletin*, 222(Part 1), 118701. <https://doi.org/10.1016/j.marpolbul.2025.118701>

**Contribution:** *Methodology, Investigation, Formal analysis, Writing – review & editing.*

6. Zesha, J. H., **Mamun, M. A.**, Islam, A. R. M. T., Habib, M. A., & Khan, R. (2025). First probabilistic radiological risk appraisal of Bay of Bengal beach sands: Explicit spatial hot-spot analysis. *Marine Pollution Bulletin*, 223, 118966. <https://doi.org/10.1016/j.marpolbul.2025.118966>

**Contribution:** *Writing – review & editing, Software, Methodology, Formal analysis, Data curation.*

7. Islam, A. R. M. T., **Mamun, M. A.**, Hasan, M., Aktar, M. N., Uddin, M. N., Siddique, M. A. B., Chowdhury, M. H., Islam, M. S., Bari, A. B. M. M., Idris, A. M., & Senapathi, V. (2024). Optimizing coastal groundwater quality predictions: A novel data mining framework with cross-validation, bootstrapping, and entropy analysis. *Journal of Contaminant Hydrology*, 269, 104480. <https://doi.org/10.1016/j.jconhyd.2024.104480>

**Contribution:** *Writing – review & editing, Methodology, Formal analysis, Data curation.*

8. **Mamun, M. A.**, Islam, A. R. M. T., Aktar, M. N., Uddin, M. N., Islam, M. S., Pal, S. C., Islam, A., Bari, A. B. M. M., Idris, A. M., & Senapathi, V. (2024). Predicting groundwater phosphate levels in coastal multi-aquifers: A geostatistical and data-driven approach. *Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2024.176024>

**Contribution:** *Writing – original draft, Software, Methodology, Formal analysis.*

## REFERENCES

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